1.1 Introduction to package qm

The `qm` package was written by Eric Majzoub, University of Missouri (email: majzoube-at-umsystem.edu). EM thanks Maxima developers Robert Dodier and Barton Willis for their contributions that have improved the package. Please report any errors or unexpected behavior by submitting an issue on the Github page for this project (https://github.com/QMeqGR/qm-maxima).

The purpose of this package is to provide computational tools for solving quantum mechanics problems in a finite-dimensional Hilbert space. It was written with students in mind as well as practitioners, and is appropriate for upper-level undergraduate quantum mechanics at the level of Townsend's *A Modern Introduction to Quantum Mechanics* and above. Using the package one may compute, for example, the hyperfine splitting in hydrogen with only a few commands (See [stationary], page 25). Numerous other examples are given throughout the manual. Additional examples are provided as wxMaxima worksheets in the `doc/examples` folder of the package.

The package is loaded with: `load(qm);`

If you use wxMaxima then issue `set_display('ascii);` after loading the `qm` package. This will allow pretty printing of the kets and bras similar to what you see in this manual. XML output can be restored with `set_display('xml);`.

1.1.1 Basic capabilities

The `qm` package provides functions and standard definitions to solve quantum mechanics problems in a finite dimensional Hilbert space. For example, one can calculate the outcome of Stern-Gerlach experiments using built-in definitions of the Sx, Sy, and Sz operators for arbitrary spin, e.g. `s={1/2, 1, 3/2, ...}`. For spin-1/2 the standard basis kets in the `x`, `y`, and `z`-basis are defined in the package as `{xp,xm}`, `{yp,ym}`, and `{zp,zm}`, respectively.
Brief list of some capabilities:

- Create kets and bras in a matrix representation with arbitrary but finite dimension.
- Create *general* ket vectors with arbitrary but finite dimension.
- Given an operator and a state ket, perform standard computations of expectation values, variance, etc.
- For time-independent Hamiltonians one can construct the time evolution operator and study, for example, magnetic resonance.
- Create kets and bras in the angular momentum `|j,m>` representation.
- Create tensor products of states to represent multi-particle systems.
- Compute Clebsch-Gordan coefficients by stepping up and down the ladder of states constructed using tensor products of `|j,m>` kets.
- Construct Bell states and compute quantities of interest for entangled systems such as density matrices.
- Perform partial traces over density matrices to compute reduced density matrices.

Let us begin with the trivial example of a spin-1/2 particle. This will illustrate how kets and bras in a matrix representation are defined within the `qm` package. A bra vector in the `z`-basis may be written as

```
<psi| = a <z+| + b <z-|.
```

The matrix representation of the bra `<psi|` will be represented in Maxima by the row vector `[a b]`, where the basis vectors are

```
<z+| = [1 0]
```

and

```
<z-| = [0 1].
```

This bra vector can be created in several ways. First, with the `mbra` command

```
mbra([a,b])
```

or by taking the quantum mechanical dagger of the corresponding ket. In a Maxima session this looks like the following. The basis kets `{zp,zm}` are transformed into bras using the `dagger` function.

```
(%i1) zp;
                                      [ 1 ]
(%o1)                                 [   ]
                                      [ 0 ]
(%i2) zm;
                                      [ 0 ]
(%o2)                                 [   ]
                                      [ 1 ]
(%i1) psi_bra:a*dagger(zp)+b*dagger(zm);
(%o1)                               [ a  b ]
(%i2) dagger(mket([a,b]));
(%o2)                               [ a  b ]
(%i3) mbra([a,b]);
(%o3)                               [ a  b ]
```

1.1.2 Kets and bras: abstract and matrix representations

There are two types of kets and bras available in the `qm` package, the first type is given by a *matrix representation*, as returned by the functions `mbra` and `mket`. `mket`s are column vectors and `mbra`s are row vectors, and their components are entered as Maxima *lists* in the `mbra` and `mket` functions. The second type of bra or ket is *abstract*; there is no matrix representation. Abstract bras and kets are entered using the `bra` and `ket` functions, while also using Maxima lists for the elements. These general kets are displayed in Dirac notation. Abstract bras and kets are used for both the `(j,m)` representation of states and also for tensor products. For example, a tensor product of two ket vectors `|a>` and `|b>` is input as `ket([a,b])` and displayed as

```
|a,b>      (general ket)
```

Note that abstract kets and bras are *assumed to be orthonormal*. These general bras and kets may be used to build arbitrarily large tensor product states.

The following examples illustrate some of the basic capabilities of the `qm` package. Here both abstract, and concrete (matrix representation) kets are shown. The last example shows how to construct an entangled Bell pair.

```
(%i1) ket([a,b])+ket([c,d]);
(%o1)                          |c, d> + |a, b>
(%i2) mket([a,b]);
                                     [ a ]
(%o2)                                [   ]
                                     [ b ]
(%i3) mbra([a,b]);
(%o3)                                [ a  b ]
(%i4) bell:(1/sqrt(2))*(ket([u,d])-ket([d,u]));
                                |u, d> - |d, u>
(%o4)                           ---------------
                                    sqrt(2)
(%i5) dagger(bell);
                                <u, d| - <d, u|
(%o5)                           ---------------
                                    sqrt(2)
```

Note that `ket([a,b])` is treated as tensor product of states `a` and `b` as shown below.

```
(%i1) braket(bra([a1,b1]),ket([a2,b2]));
(%o1)               kron_delta(a1, a2) kron_delta(b1, b2)
```

Constants that multiply kets and bras must be declared complex by the user in order for the dagger function to properly conjugate such constants. The example below illustrates this behavior.

```
(%i1) declare([a,b],complex);
(%o1)                                done
(%i2) psi:a*ket([1])+b*ket([2]);
(%o2)                             |2> b + |1> a
(%i3) psidag:dagger(psi);
(%o3)                 <2| conjugate(b) + <1| conjugate(a)
(%i4) psidag . psi;
(%o4)                   b conjugate(b) + a conjugate(a)
```

The following shows how to declare a ket with both real and complex components in the matrix representation.

```
(%i1) declare([c1,c2],complex,r,real);
(%o1)                                done
(%i2) k:mket([c1,c2,r]);
                                     [ c1 ]
                                     [    ]
(%o2)                                [ c2 ]
                                     [    ]
                                     [ r  ]
(%i3) b:dagger(k);
(%o3)                  [ conjugate(c1)  conjugate(c2)  r ]
(%i4) b . k;
                     2
(%o4)               r  + c2 conjugate(c2) + c1 conjugate(c1)
```

1.1.3 Special ket types

Some kets are difficult to work with using either the matrix representation or the general ket representation. These include tensor products of (j,m) kets used in the addition of angular momentum computations. For this reason there are a set of `tpkets` and associated `tpXX` functions defined in section `(j,m)-kets and bras`.

1.1.4 Basis sets

As described above there are three basis ket types: mkets that have a matrix representation, abstract kets that are displayed in Dirac notation, and tensor product kets (`tpkets`). Each of the three kets types can be used to construct a basis set. Valid basis sets are simply Maxima lists whose elements are one of the three ket types.

$$[b_1, b_2, b_3, \ldots]$$

Basis sets may be generated automatically using the `basis_set` shown in See [basis_set], page 8.

1.1.5 Types of spin operators: Jxx and Sxx operators

When working with kets and bras in the matrix representation, use the spin operators `Sxx`. When working with abstract kets and bras in the (j,m) representation use the operators `Jxx`. The family of `Sxx` operators are represented as matrices in Maxima, while the family of `Jxx` operators are rule based or function based.

1.1.6 The dot operator in Maxima

The dot operator "." in Maxima is used for non-commutative multiplication. In the `qm` package the dot operator is used in the representation of dyads, for example:

```
|a> . <b|
```

where `|a>` and `|b>` are arbitrary kets, and to represent brakets, with

```
<b| . |a> = <b|a>.
```

There are two important flags that control how Maxima treats the dot operator that are relevant to the `qm` package: `dotscrules` and `dotconstrules`. The first allows declared scalar multipliers to move past the dot operator, while the second allows declared constants to move past the dot operator. An example where this must be done is the following, where we use the completeness relation for the basis set given by the kets `|0>` and `|1>`. Only the last line is correctly evaluated, and required declaring `a` and `b` to be scalar, and setting `dotscrules` to `true`.

```
(%i1) V:complete(basis_set(1,[0,1]));
(%o1)                          |1> . <1| + |0> . <0|
(%i2) psi:a*ket([0])+b*ket([1]);
(%o2)                             |1> b + |0> a
(%i3) V . psi;
(%o3)                           |1> . b + |0> . a
(%i4) declare([a,b],scalar);
(%o4)                                done
(%i5) V . psi;
(%o5)                           |1> . b + |0> . a
(%i6) dotscrules:true;
(%o6)                                true
(%i7) V . psi;
(%o7)                             |1> b + |0> a
```

1.2 Functions and Variables for qm

First, let us begin with some basic constant definitions and show how to construct kets and bras.

`hbar` [Variable]

Planck's constant divided by `2*%pi`. `hbar` is not given a floating point value, but is declared to be a real number greater than zero. If the system variable `display2d_unicode` is `true` then `hbar` will be displayed as its Unicode character.

`ket` ([k_1,k_2,...]) [Function]

`ket` creates a general state ket, or tensor product, with symbols k_i representing the states. The state kets k_i are assumed to be orthonormal.

```
(%i1) k:ket([u,d]);
(%o1)                                |u, d>
(%i2) b:bra([u,d]);
(%o2)                                <u, d|
(%i3) b . k;
(%o3)                                   1
```

`ketp` (*abstract ket*) [Function]

`ketp` is a predicate function for abstract kets. It returns `true` for abstract `ket`s and `false` for anything else.

`bra` ([b_1,b_2,...]) [Function]

`bra` creates a general state bra, or tensor product, with symbols b_i representing the states. The state bras b_i are assumed to be orthonormal.

```
(%i1) k:ket([u,d]);
(%o1)                                |u, d>
(%i2) b:bra([u,d]);
(%o2)                                <u, d|
(%i3) b . k;
(%o3)                                   1
```

`brap` (*abstract bra*) [Function]

`brap` is a predicate function for abstract bras. It returns `true` for abstract `bra`s and `false` for anything else.

`mket` ([c_1,c_2,...]) [Function]

`mket` creates a *column* vector of arbitrary finite dimension. The entries c_i can be any Maxima expression. The user must `declare` any relevant constants to be complex. For a matrix representation the elements must be entered as a list in `[...]` square brackets.

```
(%i1) declare([c1,c2],complex);
(%o1)                                done
(%i2) mket([c1,c2]);
                                    [ c1 ]
(%o2)                               [    ]
                                    [ c2 ]
(%i3) properties(c1);
(%o3)          [database info, kind(c1, complex), matchdeclare]
```

`mketp` (*ket*) [Function]

`mketp` is a predicate function that checks if its input is an mket, in which case it returns `true`, else it returns `false`. `mketp` only returns `true` for the matrix representation of a ket.

```
(%i1) k:ket([a,b]);
(%o1)                               |a, b>
(%i2) mketp(k);
(%o2)                                false
(%i3) k:mket([a,b]);
                                     [ a ]
(%o3)                                [   ]
                                     [ b ]
(%i4) mketp(k);
(%o4)                                true
```

`mbra` ([c_1,c_2,...]) [Function]

`mbra` creates a *row* vector of arbitrary finite dimension. The entries c_i can be any Maxima expression. The user must `declare` any relevant constants to be complex. For a matrix representation the elements must be entered as a list in `[...]` square brackets.

```
(%i1) kill(c1,c2);
(%o1)                                done
(%i2) mbra([c1,c2]);
(%o2)                            [ c1  c2 ]
(%i3) properties(c1);
(%o3)                                 []
```

`mbrap` (*bra*) [Function]

`mbrap` is a predicate function that checks if its input is an mbra, in which case it returns `true`, else it returns `false`. `mbrap` only returns `true` for the matrix representation of a bra.

```
(%i1) b:mbra([a,b]);
(%o1)                                [ a  b ]
(%i2) mbrap(b);
(%o2)                                 true
```

Two additional functions are provided to create kets and bras in the matrix representation. These functions conveniently attempt to automatically `declare` constants as complex. For example, if a list entry is `a*sin(x)+b*cos(x)` then only `a` and `b` will be `declare`-d complex and not `x`.

`autoket` ([a_1,a_2,...]) [Function]

`autoket` takes a list [a_1,a_2,...] and returns a ket with the coefficients a_i `declare`-d complex. Simple expressions such as `a*sin(x)+b*cos(x)` are allowed and will `declare` only the coefficients as complex.

```
(%i1) autoket([a,b]);
                                     [ a ]
(%o1)                                [   ]
                                     [ b ]
(%i2) properties(a);
(%o2)           [database info, kind(a, complex), matchdeclare]
(%i1) autoket([a*sin(x),b*sin(x)]);
                                  [ a sin(x) ]
(%o1)                             [          ]
                                  [ b sin(x) ]
(%i2) properties(a);
(%o2)           [database info, kind(a, complex), matchdeclare]
```

`autobra` ([a_1,a_2,...]) [Function]

`autobra` takes a list [a_1,a_2,...] and returns a bra with the coefficients a_i `declare`-d complex. Simple expressions such as `a*sin(x)+b*cos(x)` are allowed and will `declare` only the coefficients as complex.

```
(%i1) autobra([a,b]);
(%o1)                                [ a  b ]
(%i2) properties(a);
(%o2)           [database info, kind(a, complex), matchdeclare]
(%i1) autobra([a*sin(x),b]);
(%o1)                             [ a sin(x)  b ]
(%i2) properties(a);
(%o2)           [database info, kind(a, complex), matchdeclare]
```

`mket_basis` (`n`) [Function]

`mket_basis` is a basis set generator for mkets. It takes an integer `n` and returns a set of `n` basis vectors of length `n` with `1` in the ith location of the ith vector. (This is the common z-basis matrix representation for `2j+1=n` states.)

```
(%i1) mket_basis(2);
                                  [ 1 ]  [ 0 ]
(%o1)                            [[   ], [   ]]
                                  [ 0 ]  [ 1 ]
```

1.2.1 Working with abstract basis sets

`basis_set (n,[l1,l2,...])` [Function]

The function `basis_set` takes two arguments, `n` is the number of particles, and the second argument is a list of labels of the particle states. The number of elements in the basis set is m^n, where `m` is the number of states per particle.

```
(%i1) basis_set(2,[0,1]);
(%o1)                  [|1, 1>, |1, 0>, |0, 1>, |0, 0>]
(%i2) basis_set(3,[u,d]);
(%o2) [|d, d, d>, |d, d, u>, |d, u, d>, |d, u, u>, |u, d, d>, |u, d, u>,
                                                        |u, u, d>, |u, u, u>]
```

`complete (B)` [Function]

The function `complete` generates the completeness relation for the abstract basis set `B`.

```
(%i1) B:basis_set(1,[0,1]);
(%o1)                              [|1>, |0>]
(%i2) complete(B);
(%o2)                        |1> . <1| + |0> . <0|
```

`basis_set(psi,B,[1])` [Function]

The function `coefs_psi` takes a wave function `psi` in an abstract basis set `B` and returns the coefficients of `psi` in the basis `B`. If the optional argument 1 is given the output is the expansion of `psi` in the basis `B`.

```
(%i1) declare([c1,c2],[complex,scalar])$
(%i2) B: basis_set(1,[0,1,2]);
(%o2)                           [|2>, |1>, |0>]
(%i3) psi: c0*ket([0]) + c2*ket([2]);
(%o3)                           |0> c0 + |2> c2
(%i4) coefs_psi(psi,B);
(%o4)                             [c2, 0, c0]
(%i5) coefs_psi(psi,B,1);
(%o5)                           |0> c0 + |2> c2
```

`dagger` (*vector*) [Function]

`dagger` is the quantum mechanical *dagger* function and returns the `conjugate transpose` of its input. Arbitrary constants must be `declare`-d complex for dagger to produce the conjugate.

```
(%i1) dagger(mbra([%i,2]));
                                  [ - %i ]
(%o1)                             [      ]
                                  [  2   ]
```

braket (psi,phi) [Function]

Given a bra psi and ket phi, braket returns the quantum mechanical bracket <psi|phi>. Note, braket(b,k) is equivalent to b . k where . is the Maxima non-commutative dot operator.

```
(%i1) declare([a,b,c],complex);
(%o1)                                 done
(%i2) braket(mbra([a,b,c]),mket([a,b,c]));
                                  2    2    2
(%o2)                            c  + b  + a
(%i3) braket(dagger(mket([a,b,c])),mket([a,b,c]));
(%o3)          c conjugate(c) + b conjugate(b) + a conjugate(a)
(%i4) braket(bra([a1,b1,c1]),ket([a2,b2,c2]));
(%o4)      kron_delta(a1, a2) kron_delta(b1, b2) kron_delta(c1, c2)
```

norm (psi) [Function]

Given a ket or bra psi, norm returns the square root of the quantum mechanical bracket <psi|psi>. The vector psi must always be a ket, otherwise the function will return false.

```
(%i1) declare([a,b,c],complex);
(%o1)                                 done
(%i2) norm(mket([a,b,c]));
(%o2)       sqrt(c conjugate(c) + b conjugate(b) + a conjugate(a))
```

magsqr (c) [Function]

magsqr returns conjugate(c)*c, the magnitude squared of a complex number.

```
(%i1) declare([a,b,c,d],complex);
(%o1)                                 done
(%i2) braket(mbra([a,b]),mket([c,d]));
(%o2)                               b d + a c
(%i3) P:magsqr(%);
(%o3) (b d + a c) (conjugate(b) conjugate(d) + conjugate(a) conjugate(c))
```

1.2.2 Spin-1/2 state kets and associated operators

Spin-1/2 particles are characterized by a simple 2-dimensional Hilbert space of states. It is spanned by two vectors. In the z-basis these vectors are {zp,zm}, and the basis kets in the z-basis are {xp,xm} and {yp,ym} respectively.

zp [Function]

Return the |z+> ket in the z-basis.

zm [Function]

Return the |z-> ket in the z-basis.

xp [Function]

Return the |x+> ket in the z-basis.

xm [Function]

Return the |x-> ket in the z-basis.

yp [Function]

Return the |*y*+> ket in the *z*-basis.

ym [Function]

Return the |*y*-> ket in the *z*-basis.

```
(%i1) yp;
                                      [    1    ]
                                      [ ------- ]
                                      [ sqrt(2) ]
(%o1)                                 [         ]
                                      [   %i    ]
                                      [ ------- ]
                                      [ sqrt(2) ]
(%i2) ym;
                                     [     1     ]
                                     [  -------  ]
                                     [  sqrt(2)  ]
(%o2)                                [           ]
                                     [     %i    ]
                                     [ - ------- ]
                                     [   sqrt(2) ]
(%i1) braket(dagger(xp),zp);
                                        1
(%o1)                                 -------
                                      sqrt(2)
```

Switching bases is done in the following example where a *z*-basis ket is constructed and the *x*-basis ket is computed.

```
(%i1) declare([a,b],complex);
(%o1)                                done
(%i2) psi:mket([a,b]);
                                     [ a ]
(%o2)                                [   ]
                                     [ b ]
(%i3) psi_x:'xp*braket(dagger(xp),psi)+'xm*braket(dagger(xm),psi);
                     b         a              a         b
(%o3)           (------- + -------) xp + (------- - -------) xm
                 sqrt(2)   sqrt(2)        sqrt(2)   sqrt(2)
```

1.2.3 Pauli matrices and Sz, Sx, Sy operators

sigmax [Function]

Returns the Pauli *x* matrix.

sigmay [Function]

Returns the Pauli *y* matrix.

sigmaz [Function]

Returns the Pauli *z* matrix.

`Sx` [Function]

Returns the spin-1/2 *Sx* matrix.

`Sy` [Function]

Returns the spin-1/2 *Sy* matrix.

`Sz` [Function]

Returns the spin-1/2 *Sz* matrix.

```
(%i1) sigmay;
                                  [ 0   - %i ]
(%o1)                             [          ]
                                  [ %i   0   ]
(%i2) Sy;
                                [            %i hbar ]
                                [    0     - ------- ]
                                [                2   ]
(%o2)                           [                    ]
                                [ %i hbar            ]
                                [ -------      0     ]
                                [    2               ]
```

`commutator (X,Y)` [Function]

Given two operators `X` and `Y`, return the commutator `X . Y - Y . X`.

```
(%i1) commutator(Sx,Sy);
                              [        2             ]
                              [ %i hbar              ]
                              [ --------      0      ]
                              [    2                 ]
(%o1)                         [                      ]
                              [                   2  ]
                              [             %i hbar  ]
                              [    0      - -------- ]
                              [                2     ]
```

`anticommutator (X,Y)` [Function]

Given two operators `X` and `Y`, return the commutator `X . Y + Y . X`.

```
(%i1) (1/2)*anticommutator(sigmax,sigmax);
                                    [ 1  0 ]
(%o1)                               [      ]
                                    [ 0  1 ]
```

1.2.4 SX, SY, SZ operators for any spin

`SX (s)` [Function]

`SX(s)` for spin `s` returns the matrix representation of the spin operator `Sx`. Shortcuts for spin-1/2 are `Sx,Sy,Sz`, and for spin-1 are `Sx1,Sy1,Sz1`.

SY (s) [Function]

SY(s) for spin s returns the matrix representation of the spin operator Sy. Shortcuts for spin-1/2 are Sx,Sy,Sz, and for spin-1 are Sx1,Sy1,Sz1.

SZ (s) [Function]

SZ(s) for spin s returns the matrix representation of the spin operator Sz. Shortcuts for spin-1/2 are Sx,Sy,Sz, and for spin-1 are Sx1,Sy1,Sz1.

Example:

```
(%i1) SY(1/2);
                                [            %i hbar ]
                                [    0     - ------- ]
                                [                2    ]
(%o1)                           [                     ]
                                [ %i hbar             ]
                                [ -------      0      ]
                                [    2                ]
(%i2) SX(1);
                           [           hbar            ]
                           [    0     -------     0    ]
                           [          sqrt(2)          ]
                           [                           ]
                           [  hbar              hbar   ]
(%o2)                      [ -------     0     ------- ]
                           [ sqrt(2)           sqrt(2) ]
                           [                           ]
                           [           hbar            ]
                           [    0     -------     0    ]
                           [          sqrt(2)          ]
```

1.2.5 Basis set transformations

Given a matrix representation of an operator in terms of mkets one may transform from one mket basis to another.

basis_set_p (B) [Function]

The predicate function basis_set_p takes as an argument a basis set [b_1,b_2,...] enclosed in square brackets, where each b_i is true for the predicate function mketp.

```
(%i1) basis_set_p([zp,zm]);
(%o1)                                true
```

mtrans (B_1,B_2) [Function]

The function mtrans returns the matrix of inner products of the two bases B_1 and B_2. The bases must be of the same dimension.

```
(%i1) mtrans([zp,zm],[yp,ym]);
                              [    1         1     ]
                              [ -------   -------  ]
                              [ sqrt(2)   sqrt(2)  ]
(%o1)                         [                    ]
                              [   %i        %i     ]
                              [ -------  - ------- ]
                              [ sqrt(2)    sqrt(2) ]
```

`op_trans (A,B1,B2)` [Function]

The function `op_trans` returns the matrix representation of operator `A` in basis B_2. The operator `A` must be given in the basis B_1.

```
(%i1) op_trans(Sy,[zp,zm],[yp,ym]);
                                  [ hbar         ]
                                  [ ----    0    ]
                                  [  2           ]
(%o1)                             [               ]
                                  [         hbar  ]
                                  [  0    - ----  ]
                                  [           2   ]
```

1.2.6 Expectation value and variance

`expect (O,psi)` [Function]

Computes the quantum mechanical expectation value of the operator `O` in state `psi`, `<psi|O|psi>`.

```
(%i1) expect(Sy,xp+ym),expand;
(%o1)                                - hbar
```

`qm_variance (O,psi)` [Function]

Computes the quantum mechanical variance of the operator `O` in state `psi`, `sqrt(<psi|O^2|psi> - <psi|O|psi>^2)`.

```
(%i1) qm_variance(Sy,xp+ym),expand;
                                     %i hbar
(%o1)                                -------
                                        2
```

1.2.7 Angular momentum and ladder operators in the matrix representation

`SP (s)` [Function]

`SP` is the raising ladder operator S_+ for spin `s`.

`SM (s)` [Function]

`SM` is the raising ladder operator S_- for spin `s`.

Examples of the ladder operators:

```
(%i1) SP(1);
                        [ 0  sqrt(2) hbar       0       ]
                        [                               ]
(%o1)                   [ 0       0        sqrt(2) hbar ]
                        [                               ]
                        [ 0       0             0       ]
(%i2) SM(1);
                        [      0             0        0 ]
                        [                             ]
(%o2)                   [ sqrt(2) hbar       0        0 ]
                        [                             ]
                        [      0        sqrt(2) hbar  0 ]
```

1.2.8 Rotation operators

`RX (s,t)` [Function]

`RX(s)` for spin `s` returns the matrix representation of the rotation operator `Rx` for rotation through angle `t`.

`RY (s,t)` [Function]

`RY(s)` for spin `s` returns the matrix representation of the rotation operator `Ry` for rotation through angle `t`.

`RZ (s,t)` [Function]

`RZ(s)` for spin `s` returns the matrix representation of the rotation operator `Rz` for rotation through angle `t`.

```
(%i1) RY(1,t);
                      [ cos(t) + 1    sin(t)   1 - cos(t) ]
                      [ ----------  - -------  ---------- ]
                      [     2         sqrt(2)      2      ]
                      [                                   ]
                      [  sin(t)                   sin(t)  ]
(%o1)                 [  -------     cos(t)     - -------  ]
                      [  sqrt(2)                  sqrt(2)  ]
                      [                                   ]
                      [ 1 - cos(t)   sin(t)    cos(t) + 1 ]
                      [ ----------   -------   ---------- ]
                      [     2        sqrt(2)       2      ]
```

1.2.9 Time-evolution operator

`U (H,t)` [Function]

`U(H,t)` is the time evolution operator for Hamiltonian `H`. It is defined as the matrix exponential `matrixexp(-%i*H*t/hbar)`.

```
(%i1) assume(w > 0);
(%o1)                                [w > 0]
(%i2) U(w*Sy,t);
                             [     t w          t w  ]
                             [ cos(---)  - sin(---) ]
                             [      2            2   ]
(%o2)                        [                       ]
                             [     t w         t w   ]
                             [ sin(---)   cos(---)   ]
                             [      2           2    ]
```

1.3 Angular momentum representation of kets and bras

1.3.1 Matrix representation of (j,m)-kets and bras

The matrix representation of kets and bras in the `qm` package are represented in the `z`-basis. To create a matrix representation of of a ket or bra in the (j,m)-basis one uses the `spin_mket` and `spin_mbra` functions.

`spin_mket` (s,m_s,[*1,2,3*]) [Function]

`spin_mket` returns a ket in the `z`-basis for spin `s` and z-projection $\mathtt{m}_\mathtt{s}$, for axis 1=X, 2=Y, 3=Z.

`spin_mbra` (s,m_s,[*1,2,3*]) [Function]

`spin_mbra` returns a bra in the `z`-basis for spin `s` and z-projection $\mathtt{m}_\mathtt{s}$, for axis 1=X, 2=Y, 3=Z.

```
(%i1) spin_mbra(3/2,1/2,2);
                     [ sqrt(3)     %i    1      sqrt(3) %i ]
(%o1)                [ -------  - ----  ----  - ---------- ]
                     [   3/2       3/2   3/2        3/2    ]
                     [  2         2     2          2       ]
```

1.3.2 Angular momentum (j,m)-kets and bras

To create kets and bras in the |*j,m*> representation you use the abstract `ket` and `bra` functions with `j,m` as arguments, as in `ket([j,m])` and `bra([j,m])`.

```
(%i1) bra([3/2,1/2]);
                                      3  1
(%o1)                                <-, -|
                                      2  2
(%i2) ket([3/2,1/2]);
                                      3  1
(%o2)                                |-, ->
                                      2  2
```

Some convenience functions for making the kets are the following:

`jmtop` (j) [Function]

`jmtop` creates a (j,m)-ket with `m=j`.

```
(%i1) jmtop(3/2);
                                      3  3
(%o1)                                |-, ->
                                      2  2
```

`jmbot (j)` [Function]

`jmbot` creates a (j,m)-ket with `m=-j`.

```
(%i1) jmbot(3/2);
                                     3    3
(%o1)                               |-, - ->
                                     2    2
```

`jmket (j,m)` [Function]

`jmket` creates a (j,m)-ket.

```
(%i1) jmket(3/2,1/2);
                                      3  1
(%o1)                                |-, ->
                                      2  2
```

`jmketp` (*jmket*) [Function]

`jmketp` checks to see that the ket has an `m`-value that is in the set `{-j,-j+1,...,+j}`.

```
(%i1) jmketp(ket([j,m]));
(%o1)                                false
(%i2) jmketp(ket([3/2,1/2]));
(%o2)                                true
```

`jmbrap` (*jmbra*) [Function]

`jmbrap` checks to see that the bra has an `m`-value that is in the set `{-j,-j+1,...,+j}`.

`jmcheck (j,m)` [Function]

`jmcheck` checks to see that m is one of {-j, ..., +j}.

```
(%i1) jmcheck(3/2,1/2);
(%o1)                                true
```

`Jp` (*jmket*) [Function]

`Jp` is the J_+ operator. It takes a `jmket jmket(j,m)` and returns `sqrt(j*(j+1)-m*(m+1))*hbar*jmket(j,m+1)`.

`Jm` (*jmket*) [Function]

`Jm` is the J_- operator. It takes a `jmket jmket(j,m)` and returns `sqrt(j*(j+1)-m*(m-1))*hbar*jmket(j,m-1)`.

`Jsqr` (*jmket*) [Function]

`Jsqr` is the J^2 operator. It takes a `jmket jmket(j,m)` and returns `j*(j+1)*hbar`^{2}`*jmket(j,m)`.

`Jz` (*jmket*) [Function]

`Jz` is the J_z operator. It takes a `jmket jmket(j,m)` and returns `m*hbar*jmket(j,m)`.

These functions are illustrated below.

```
(%i1) k:ket([j,m]);
(%o1)                                |j, m>
(%i2) Jp(k);
(%o2)            hbar |j, m + 1> sqrt(j (j + 1) - m (m + 1))
(%i3) Jm(k);
(%o3)            hbar |j, m - 1> sqrt(j (j + 1) - (m - 1) m)
(%i4) Jsqr(k);
                                2
(%o4)                       hbar  j (j + 1) |j, m>
(%i5) Jz(k);
(%o5)                            hbar |j, m> m
```

1.3.3 Addition of angular momentum in the (j,m)-representation

Addition of angular momentum calculations can be performed in the (j,m)-representation using the function definitions below. The internal representation of kets and bras for this purpose is the following. Given kets `|j1,m1>` and `|j2,m2>` a tensor product of (j,m)-kets is instantiated as:

```
tpket(1,|j1,m1>,|j2,m2>)
```

and the corresponding bra is instantiated as:

```
tpbra(1,<j1,m1|,<j2,m2|)
```

where the factor of 1 is the multiplicative factor of the tensor product. We call this the *common factor* (cf) of the tensor product. The general form of a tensor product in the (j,m) representation is:

```
tpket( cf, |j1,m1>, |j2,m2> ).
```

`tpket` (*jmket1,jmket2*) [Function]

`tpket` instantiates a tensor product of two (j,m)-kets.

```
(%i1) tpket(ket([3/2,1/2]),ket([1/2,1/2]));
                                3  1     1  1
(%o1)                          |-, -> ⊗ |-, ->
                                2  2     2  2
```

`tpbra` (*jmbra1,jmbra2*) [Function]

`tpbra` instantiates a tensor product of two (j,m)-bras.

```
(%i1) tpbra(bra([3/2,1/2]),bra([1/2,1/2]));
                                3  1     1  1
(%o1)                          <-, -| ⊗ <-, -|
                                2  2     2  2
```

`tpbraket` (*tpbra,tpket*) [Function]

`tpbraket` returns the bracket of a `tpbra` and a `tpket`.

```
(%i1) k:tpket(jmtop(1),jmbot(1));
(%o1)                          |1, 1> ⊗ |1, - 1>
(%i2) K:Jtsqr(k);
                     2                             2
(%o2)         2 hbar  |1, 1> ⊗ |1, - 1> + 2 hbar  |1, 0> ⊗ |1, 0>
(%i3) B:tpdagger(k);
(%o3)                          <1, 1| ⊗ <1, - 1|
(%i4) tpbraket(B,K);
                                        2
(%o4)                              2 hbar
```

`tpdagger` (*tpket or tpbra*) [Function]

`tpdagger` takes the quantum mechanical dagger of a `tpket` or `tpbra`.

```
(%i1) k1:tpket(ket([1/2,1/2]),ket([1/2,-1/2]));
                                1  1     1    1
(%o1)                          |-, -> ⊗ |-, - ->
                                2  2     2    2
(%i2) tpdagger(k1);
                                1  1     1    1
(%o2)                          <-, -| ⊗ <-, - -|
                                2  2     2    2
```

`J1z` (*tpket*) [Function]

`J1z` returns the tensor product of a tpket with `Jz` acting on the first ket.

`J2z` (*tpket*) [Function]

`J2z` returns the tensor product of a tpket with `Jz` acting on the second ket.

```
(%i1) k:tpket(ket([3/2,3/2]),ket([1/2,1/2]));
                                 3  3     1  1
(%o1)                           |-, -> ⊗ |-, ->
                                 2  2     2  2
(%i2) J1z(k);
                                    3  3     1  1
                              3 hbar |-, -> ⊗ |-, ->
                                    2  2     2  2
(%o2)                         ----------------------
                                       2
(%i3) J2z(k);
                                   3  3     1  1
                             hbar |-, -> ⊗ |-, ->
                                   2  2     2  2
(%o3)                        --------------------
                                       2
```

`Jtz` (*tpket*) [Function]

`Jtz` is the total z-projection of spin operator acting on a tpket and returning (J_{1z}+J_{2z}).

```
(%i1) k:tpket(ket([3/2,3/2]),ket([1/2,1/2]));
                                  3  3     1  1
(%o1)                            |-, -> ⊗ |-, ->
                                  2  2     2  2
(%i2) Jtz(k);
                                     3  3     1  1
(%o2)                       2 hbar |-, -> ⊗ |-, ->
                                     2  2     2  2
```

`J1sqr` (*tpket*) [Function]

`J1sqr` returns `Jsqr` for the first ket of a tpket.

`J2sqr` (*tpket*) [Function]

`J2sqr` returns `Jsqr` for the second ket of a tpket.

`J1p` (*tpket*) [Function]

`J1p` returns J_+ for the first ket of a tpket.

`J2p` (*tpket*) [Function]

`J2p` returns J_+ for the second ket of a tpket.

`Jtp` (*tpket*) [Function]

`Jtp` returns $(J_{1+}+J_{2+})$ for the tpket.

`J1m` (*tpket*) [Function]

`J1m` returns J_- for the first ket of a tpket.

`J2m` (*tpket*) [Function]

`J2m` returns J_- for the second ket of a tpket.

`Jtm` (*tpket*) [Function]

`Jtm` returns $(J_{1-}+J_{2-})$ for the tpket.

`J1p2m` (*tpket*) [Function]

`J1p2m` returns $(J_{1+}J_{2-})$ for the tpket.

```
(%i1) k:tpket(ket([3/2,1/2]),ket([1/2,1/2]));
                                  3  1     1  1
(%o1)                            |-, -> ⊗ |-, ->
                                  2  2     2  2
(%i2) b:tpdagger(k);
                                  3  1     1  1
(%o2)                            <-, -| ⊗ <-, -|
                                  2  2     2  2
(%i3) J1p2m(k);
                                     2  3  3     1    1
(%o3)                   sqrt(3) hbar  |-, -> ⊗ |-, - ->
                                     2  2     2    2
(%i4) J1m2p(k);
(%o4)                                0
```

`J1m2p` (*tpket*) [Function]

`J1m2p` returns $J_{1-}J_{2+}$ for the tpket.

`J1zJ2z` (*tpket*) [Function]

`J1zJ2z` returns $J_{1z}J_{2z}$ for the tpket.

`Jtsqr` (*tpket*) [Function]

`Jtsqr` returns $J_1{}^2+J_2{}^2+ J_{1+}J_{2-}+J_{1-}J_{2+}+J_{1z}J_{2z}$ for the tpket.

```
(%i1) k:tpket(ket([3/2,-1/2]),ket([1/2,1/2]));
                                3    1     1  1
(%o1)                          |-, - -> ⊗ |-, ->
                                2    2     2  2
(%i2) B:tpdagger(k);
                                3    1     1  1
(%o2)                          <-, - -| ⊗ <-, -|
                                2    2     2  2
(%i3) K2:Jtsqr(k);
                    2  3  1     1    1          2  3    1     1  1
(%o3)        2 hbar  |-, -> ⊗ |-, - -> + 4 hbar  |-, - -> ⊗ |-, ->
                     2  2     2    2             2    2     2  2
(%i4) tpbraket(B,K2);
                                          2
(%o4)                              4 hbar
```

`get_j` (`q`) [Function]

`get_j` is a convenience function that computes `j` from `j(j+1)=q` where `q` is a rational number. This function is useful after using the function `Jtsqr`.

```
(%i1) get_j(15/4);
                                      3
(%o1)                            j = -
                                      2
```

1.3.4 Example computations

For the first example, let us see how to determine the total spin state `|j,m>` of the two-particle state `|1/2,1/2;1,1>`.

```
(%i1) k:tpket(jmtop(1/2),jmtop(1));
                                 1  1
(%o1)                           |-, -> ⊗ |1, 1>
                                 2  2
(%i2) Jtsqr(k);
                                  2  1  1
                           15 hbar  |-, -> ⊗ |1, 1>
                                    2  2
(%o2)                      ------------------------
                                     4
(%i3) get_j(15/4);
                                        3
(%o3)                                j = -
                                        2
```

This is an eigenket of `Jtsqr`, thus `|3/2,3/2> = |1/2,1/2;1,1>`, and it is also the top state. One can now apply the lowering operator to find the other states: `|3/2,1/2>`, `|3/2,-1/2>`, and `|3/2,-3/2>`.

```
(%i1) k:tpket(jmtop(1/2),jmtop(1));
                                 1  1
(%o1)                           |-, -> ⊗ |1, 1>
                                 2  2
(%i2) k2:Jtm(k);
                           1  1                    1    1
(%o2)        sqrt(2) hbar |-, -> ⊗ |1, 0> + hbar |-, - -> ⊗ |1, 1>
                           2  2                    2    2
(%i3) k3:Jtm(k2);
                                  1  1                      1    1
(%o3) sqrt(2) hbar (sqrt(2) hbar |-, -> ⊗ |1, - 1> + hbar |-, - -> ⊗ |1, 0>)
                                  2  2                      2    2
                                                             2  1    1
                                                  + sqrt(2) hbar  |-, - -> ⊗ |1, 0>
                                                             2    2
(%i4) k4:Jtm(k3);
                                    3  1    1
(%o4)                        6 hbar  |-, - -> ⊗ |1, - 1>
                                     2    2
```

In the example below we calculate the Clebsch-Gordan coefficients of the two-particle state with two spin-1/2 particles. We begin by defining the top rung of the ladder and stepping down. To calculate the coefficients one first creates the tensor product top state, and computes the values for the total angular momentum `|J,M>` at the top of the ladder `M=J`. For the first step down the ladder one computes `Jm |J,J>`, which must be equal to `Jtm |j1,j1;j2,j2>`. We then divide by the coefficient of `|J,J-1>`. This gives first set of coefficients, which in general are given by `<J,M||j1,m1;j2,m2>`. One then continues down the ladder to compute the others.

```
(%i1) top:tpket(jmtop(1/2),jmtop(1/2));
                                1  1     1  1
(%o1)                          |-, -> ⊗ |-, ->
                                2  2     2  2
(%i2) Jtsqr(top);
                                  2  1  1     1  1
(%o2)                        2 hbar  |-, -> ⊗ |-, ->
                                   2  2     2  2
(%i3) get_j(2);
(%o3)                               j = 1
(%i4) Jtz(top);
                                 1  1     1  1
(%o4)                         hbar |-, -> ⊗ |-, ->
                                 2  2     2  2
(%i5) JMtop:ket([1,1]);
(%o5)                               |1, 1>
(%i6) mid:Jtm(top);
                      1  1     1    1          1    1     1  1
(%o6)           hbar |-, -> ⊗ |-, - -> + hbar |-, - -> ⊗ |-, ->
                      2  2     2    2          2    2     2  2
(%i7) Jm(JMtop);
(%o7)                          sqrt(2) hbar |1, 0>
(%i8) mid: 1/(sqrt(2)*hbar)*mid,expand;
                      1  1     1    1     1    1     1  1
                     |-, -> ⊗ |-, - ->   |-, - -> ⊗ |-, ->
                      2  2     2    2     2    2     2  2
(%o8)                ----------------- + -----------------
                           sqrt(2)             sqrt(2)
(%i9) bot:Jtm(mid);
                                    1    1     1    1
(%o9)                 sqrt(2) hbar |-, - -> ⊗ |-, - ->
                                    2    2     2    2
(%i10) Jm(ket([1,0]));
(%o10)                         sqrt(2) hbar |1, - 1>
(%i11) bot: 1/(sqrt(2)*hbar)*bot,expand;
                                  1    1     1    1
(%o11)                           |-, - -> ⊗ |-, - ->
                                  2    2     2    2
```

1.4 General tensor products

Tensor products are represented as lists in the `qm` package. The ket tensor product `|z+,z+>` can be represented as `ket([u,d])`, for example, and the bra tensor product `<a,b|` is represented as `bra([a,b])` for states `a` and `b`. For a tensor product where the identity is one of the elements of the product, substitute the string `Id` in the ket or bra at the desired location. See the examples below for the use of the identity in tensor products.

Examples below show how to create abstract tensor products that contain the identity element `Id` and how to take the bracket of these tensor products.

```
(%i1) K:ket([a1,b1]);
(%o1)                              |a1, b1>
(%i2) B:bra([a2,b2]);
(%o2)                              <a2, b2|
(%i3) braket(B,K);
(%o3)                kron_delta(a1, a2) kron_delta(b1, b2)
(%i1) bra([a1,Id,c1]) . ket([a2,b2,c2]);
(%o1)          |-, b2, -> kron_delta(a1, a2) kron_delta(c1, c2)
(%i2) bra([a1,b1,c1]) . ket([Id,b2,c2]);
(%o2)          <a1, -, -| kron_delta(b1, b2) kron_delta(c1, c2)
```

In the next example we construct the state function for an entangled Bell pair, construct the density matrix, and then trace over the first particle to obtain the density submatrix for particle 2.

```
(%i1) bell:(1/sqrt(2))*(ket([u,d])-ket([d,u]));
                                |u, d> - |d, u>
(%o1)                           ---------------
                                    sqrt(2)
(%i2) rho:bell . dagger(bell);
      |u, d> . <u, d| - |u, d> . <d, u| - |d, u> . <u, d| + |d, u> . <d, u|
(%o2) ------------------------------------------------------------------------
                                       2
(%i3) assume(not equal(u,d));
(%o3)                           [notequal(u, d)]
(%i4) trace1:bra([u,Id]) . rho . ket([u,Id])+bra([d,Id]) . rho . ket([d,Id]);
                        |-, u> . <-, u|   |-, d> . <-, d|
(%o4)                   --------------- + ---------------
                              2                 2
```

One can also construct the density matrix using the function `matrep`.

`matrep (A,B)` [Function]

Given an abstract representation of an operator, e.g. `A = |a> . <b| + |b> . <a|`, the function `matrep` takes the operator `A` and basis set `B` and constructs the matrix representation of `A`. NOTE: if there are symbolic constants as coefficients in the abstract representation they must be `declared`d as scalar for the simplification rules to work properly with the non-commutative "." operator.

```
(%i1) bell:(1/sqrt(2))*(ket([1,0])-ket([0,1]));
                                 |1, 0> - |0, 1>
(%o1)                            ---------------
                                     sqrt(2)
(%i2) rho:bell . dagger(bell);
      |1, 0> . <1, 0| - |1, 0> . <0, 1| - |0, 1> . <1, 0| + |0, 1> . <0, 1|
(%o2) ------------------------------------------------------------------------
                                          2
(%i3) B:[ket([1,1]),ket([1,0]),ket([0,1]),ket([0,0])];
(%o3)                  [|1, 1>, |1, 0>, |0, 1>, |0, 0>]
(%i4) matrep(rho,B);
                               [ 0   0    0   0 ]
                               [                ]
                               [     1     1    ]
                               [ 0   -   - -  0 ]
                               [     2     2    ]
(%o4)                          [                ]
                               [      1   1     ]
                               [ 0  - -   -   0 ]
                               [      2   2     ]
                               [                ]
                               [ 0   0    0   0 ]
(%i5) declare([a,b],scalar);
(%o5)                                done
(%i6) O:a*ket([1]) . bra([0])+b*ket([0]) . bra([1]);
(%o6)                     (|0> . <1|) b + (|1> . <0|) a
(%i7) B:[ket([1]),ket([0])];
(%o7)                             [|1>, |0>]
(%i8) matrep(O,B);
                                  [ 0  a ]
(%o8)                             [      ]
                                  [ b  0 ]
```

1.4.1 Example calculation of matrix elements

Let us see how to compute the matrix elements of the operator `(J1z-J1z)` in the z-basis for two spin-1/2 particles. First, we define the four basis kets of the form $|j_1,m_1;j_2,m_2\rangle$. Next we define the Hamiltonian and then use the function `matrep`.

```
(%i1) b1:tpket(ket([1/2,1/2]),ket([1/2,1/2]));
                                1  1     1  1
(%o1)                          |-, -> ⊗ |-, ->
                                2  2     2  2
(%i2) b2:tpket(ket([1/2,1/2]),ket([1/2,-1/2]));
                                1  1     1    1
(%o2)                          |-, -> ⊗ |-, - ->
                                2  2     2    2
(%i3) b3:tpket(ket([1/2,-1/2]),ket([1/2,1/2]));
                                1    1     1  1
(%o3)                          |-, - -> ⊗ |-, ->
                                2    2     2  2
(%i4) b4:tpket(ket([1/2,-1/2]),ket([1/2,-1/2]));
                                1    1     1    1
(%o4)                          |-, - -> ⊗ |-, - ->
                                2    2     2    2
(%i5) B:[b1,b2,b3,b4];
       1  1     1  1    1  1     1    1    1    1     1  1
(%o5) [|-, -> ⊗ |-, ->, |-, -> ⊗ |-, - ->, |-, - -> ⊗ |-, ->,
       2  2     2  2    2  2     2    2    2    2     2  2

                                                      1    1     1    1
                                                     |-, - -> ⊗ |-, - ->]
                                                      2    2     2    2
(%i6) H1:omega*(J1z-J2z);
(%o6)                          (J1z - J2z) omega
(%i7) declare(omega,scalar);
(%o7)                               done
(%i8) matrep(H1,B);
                     [ 0      0            0        0 ]
                     [                               ]
                     [ 0  hbar omega       0        0 ]
(%o8)                [                               ]
                     [ 0      0       - hbar omega  0 ]
                     [                               ]
                     [ 0      0            0        0 ]
```

1.4.2 Stationary states from a Hamiltonian

`stationary (evals,evecs,basis)` [Function]

The function `stationary` takes the output of the `eigenvectors` command and a basis set and constructs the stationary states from the basis used to construct the matrix representation of the Hamiltonian.

Example:

The hyperfine splitting in the hydrogen atom is due to the spin-spin interaction of the electron and the proton. The Hamiltonian is 2*A/hbar^2 * ($S_1 \bullet S_2$). Let's calculate the energy levels and the stationary states.

```
(%i1) declare(A,scalar);
(%o1)                                done
(%i2) Hhyp:(A/hbar^2)*(J1p2m+J1m2p+2*J1zJ2z);
                          A (2 J1zJ2z + J1p2m + J1m2p)
(%o2)                     ---------------------------
                                       2
                                    hbar
(%i3) Hmat:matrep(Hhyp,bj1212);
                                [ A              ]
                                [ -   0    0   0 ]
                                [ 2              ]
                                [                ]
                                [      A         ]
                                [ 0  - -   A   0 ]
                                [      2         ]
(%o3)                           [                ]
                                [           A    ]
                                [ 0   A   - -  0 ]
                                [           2    ]
                                [                ]
                                [              A ]
                                [ 0   0    0   - ]
                                [              2 ]
(%i4) [evals,evecs]:eigenvectors(Hmat);
           3 A  A
(%o4) [[[- ---, -], [1, 3]], [[0, 1, - 1, 0]],
            2   2
                                    [[1, 0, 0, 0], [0, 1, 1, 0], [0, 0, 0, 1]]]]
(%i5) states:stationary(evals,evecs,bj1212);
         1  1     1    1     1    1     1  1    1  1     1  1
(%o5) [|-, -> ⊗ |-, - -> - |-, - -> ⊗ |-, ->, |-, -> ⊗ |-, ->,
         2  2     2    2     2    2     2  2    2  2     2  2
                      1  1     1    1     1    1     1  1    1    1     1    1
                     |-, -> ⊗ |-, - -> + |-, - -> ⊗ |-, ->, |-, - -> ⊗ |-, - ->]
                      2  2     2    2     2    2     2  2    2    2     2    2
(%i6) Jtz(states[1]);
(%o6)                                  0
```

1.4.3 Matrix trace functions

`qm_mtrace` (*matrix*) [Function]

The function `qm_mtrace` is the usual matrix trace; it takes a square matrix and returns the sum of the diagonal components.

`qm_atrace (A,B)` [Function]

The function `qm_atrace` takes an abstract operator `A` and a basis `B` and attempts to compute the matrix representation using the `matrep` function. If successful it will return the matrix trace of the resulting matrix.

```
(%i1) B:[ket([1]),ket([0])];
(%o1)                              [|1>, |0>]
(%i2) declare(c,scalar);
(%o2)                                 done
(%i3) A:c*ket([1]) . bra([1]);
(%o3)                             (|1> . <1|) c
(%i4) matrep(A,B);
                                    [ c  0 ]
(%o4)                               [      ]
                                    [ 0  0 ]
(%i5) qm_atrace(A,B);
(%o5)                                  c
(%i6) bell:(1/sqrt(2))*(ket([1,0])-ket([0,1]));
                                 |1, 0> - |0, 1>
(%o6)                            ---------------
                                    sqrt(2)
(%i7) rho:bell . dagger(bell);
      |1, 0> . <1, 0| - |1, 0> . <0, 1| - |0, 1> . <1, 0| + |0, 1> . <0, 1|
(%o7) ------------------------------------------------------------------------
                                        2
(%i8) trace1:bra([1,Id]) . rho . ket([1,Id])+bra([0,Id]) . rho . ket([0,Id]);
                        |-, 1> . <-, 1|   |-, 0> . <-, 0|
(%o8)                   --------------- + ---------------
                              2                 2
(%i9) B:[ket([Id,1]),ket([Id,0])];
(%o9)                           [|Id, 1>, |Id, 0>]
(%i10) matrep(trace1,B);
                                    [ 1    ]
                                    [ -  0 ]
                                    [ 2    ]
(%o10)                              [      ]
                                    [    1 ]
                                    [ 0  - ]
                                    [    2 ]
```

1.5 Quantum harmonic oscillator

The `qm` package can perform simple quantum harmonic oscillator calculations involving the ladder operators a^+ and a^-. These are referred to in the package as `ap` and `am` respectively. For computations with arbitrary states to work you must `declare` the harmonic oscillator state, say `n`, to be both `scalar` and `integer`, as shown in the examples below.

ap [Function]

ap is the raising operator a^+ for quantum harmonic oscillator states.

am [Function]

a is the lowering operator a^- for quantum harmonic oscillator states.

A common problem is to compute the 1st order change in energy of a state due to a perturbation of the harmonic potential, say an additional factor V(x) = x^2 + g*x^4 for small g. This example is performed below, ignoring any physical constants in the problem.

```
(%i1) declare(n,integer,n,scalar);
(%o1)                                done
(%i2) ap . ket([n]);
(%o2)                          sqrt(n + 1) |n + 1>
(%i3) am . ket([n]);
(%o3)                            |n - 1> sqrt(n)
(%i4) bra([n]) . (ap+am)^^4 . ket([n]);
                                      2
(%o4)                           6 n  + 6 n + 3
```

Another package that handles quantum mechanical operators is operator_algebra written by Barton Willis.

1.6 General analytic calculations

To perform more abstract calculations we define two new operator types: qop and eigenop. A qop is a quantum operator whose actions on kets are undefined, outside of yielding an abstract matrix element. An eigenop is a quantum operator whose action on kets |i> yields E_i |i>. These operators must be declared using the following two functions.

declare_qop (*symbol*) [Function]

The function declare_qop declares a symbol to be a quantum operator. The action of a qop on kets is undefined, however qops are allowed to enter summations and matrix elements are displayed if kets are represented in a numeral basis, i.e., <i| . H . |j> = H_{ij}.

```
(%i1) declare_qop(H);
(%o1)                                 qop
(%i2) bra([i]) . H . ket([j]);
(%o2)                                 H
                                       i, j
```

declare_eigenop (*symbol*) [Function]

The function declare_eigenop declares a symbol to be a quantum operator (qop) and also an eigenop. The action of a eigenop on kets is defined by the ket symbol, e.g. H . |i> = E_i |i>.

```
(%i1) declare_eigenop(H);
(%o1)                                 eigenop
(%i2) H . ket([j]);
(%o2)                                 E  |j>
                                       j
(%i3) bra([i]) . H . ket([j]);
(%o3)                        kron_delta(i, j) E
                                                j
```

Uop (H,t) [Function]

`Uop(H,t)` is the time evolution operator for eigenop Hamiltonian `H`. It is defined as the matrix exponential `matrixexp(-%i*H*t/hbar)`.

1.7 Example: Two-state systems

In this section we recast the Schrodinger equation in terms of coefficient functions generally, and then use the resulting equations to examine a simple two-state system.

```
(%i1) declare_qop(H);
(%o1)                                  qop
(%i2) declare(c,complex);
(%o2)                                 done
(%i3) depends(c,t);
(%o3)                                [c(t)]
(%i4) psi:sum(c[i]*ket([i]),i,1,N);
                                    N
                                   ____
                                   \
(%o4)                               >    c  |i>
                                   /      i
                                   ----
                                   i = 1
(%i5) schro:%i*diff(psi,t) = H . psi;
                       N                       N
                      ____                    ____
                      \      d                \
(%o5)             %i   >    (-- (c )) |i> =   >    H . (c  |i>)
                      /      dt   i          /           i
                      ----                   ----
                      i = 1                  i = 1
(%i6) bra([j]) . schro;
                                           N
                                          ____
                                          \
                             d            \
(%o6)                    %i (-- (c )) =   >    c  H
                             dt   j       /      i  j, i
                                          ----
                                          i = 1
```

This is the general formula (with hbar=1) found in textbooks. Now let us specialize to a two-state system and solve it. We will specify the matrix elements for a two-state system with transitions between two states of the same energy.

```
(%i1) declare_qop(H);
(%o1)                                qop
(%i2) psi2:c[1](t)*ket([1])+c[2](t)*ket([2]);
(%o2)                       |2> c (t) + |1> c (t)
                                 2           1
(%i3) schro2:%i*diff(psi2,t) = H . psi2;
              d                  d
(%o3) %i (|2> (-- (c (t))) + |1> (-- (c (t)))) =
              dt   2             dt   1
                                                H . (|2> c (t)) + H . (|1> c (t))
                                                          2                 1
(%i4) eq1:bra([1]) . schro2;
                     d
(%o4)            %i (-- (c (t))) = H     c (t) + H     c (t)
                     dt   1         1, 2  2       1, 1  1
(%i5) eq2:bra([2]) . schro2;
                     d
(%o5)            %i (-- (c (t))) = H     c (t) + H     c (t)
                     dt   2         2, 2  2       2, 1  1
(%i6) H:matrix([E0,-A],[-A,E0]);
                               [ E0   - A ]
(%o6)                          [          ]
                               [ - A  E0  ]
(%i7) assume(A > 0,E0 > 0);
(%o7)                          [A > 0, E0 > 0]
(%i8) eq1:''eq1;
                       d
(%o8)              %i (-- (c (t))) = E0 c (t) - A c (t)
                       dt   1            1         2
(%i9) eq2:''eq2;
                       d
(%o9)              %i (-- (c (t))) = E0 c (t) - A c (t)
                       dt   2            2         1
(%i10) soln:desolve([eq1,eq2],[c[1](t),c[2](t)])$
(%i11) add:fullsimp(soln[1]+soln[2]);
                              %i A t - %i E0 t
(%o11)       c (t) + c (t) = %e                (c (0) + c (0))
              2       1                          2       1
(%i12) sub:fullsimp(soln[1]-soln[2]);
                              - %i E0 t - %i A t
(%o12)     c (t) - c (t) = - %e                  (c (0) - c (0))
            1       2                            2       1
```

1.8 Pre-defined quantities

There are some pre-defined quantities in the file `predef.mac` that may be convenient for the user. These include Bell states, and some basis sets that are tedious to input.

```
bell1: 1/sqrt(2)*(ket([1,0])-ket([0,1]));
bell2: 1/sqrt(2)*(ket([1,0])+ket([0,1]));
bell3: 1/sqrt(2)*(ket([0,0])+ket([1,1]));
bell4: 1/sqrt(2)*(ket([0,0])-ket([1,1]));

ghz1: 1/sqrt(2)*(ket([0,0,0])-ket([1,1,1]));
ghz2: 1/sqrt(2)*(ket([0,0,0])+ket([1,1,1]));


/* pre-defined tpket bases */
bj1212: two spin-1/2 particles
bj112: one spin-1 and one spin-1/2 particle
bj11: two spin-1 particles
```

Appendix A Function and Variable index

A

am 28
anticommutator 11
ap 28
autobra 7
autoket 7

B

basis_set 8
basis_set(psi,B,[1]) 8
basis_set_p 12
bra 5
braket 9
brap 6

C

commutator 11
complete 8

D

dagger 8
declare_eigenop 28
declare_qop 28

E

expect 13

G

get_j 20

J

J1m 19
J1m2p 20
J1p 19
J1p2m 19
J1sqr 19
J1z 18
J1zJ2z 20
J2m 19
J2p 19
J2sqr 19
J2z 18
Jm 16
jmbot 16
jmbrap 16
jmcheck 16
jmket 16
jmketp 16
jmtop 15
Jp 16
Jsqr 16
Jtm 19
Jtp 19
Jtsqr 20
Jtz 18
Jz 16

K

ket 5
ketp 5

M

magsqr 9
matrep 23
mbra 6
mbrap 7
mket 6
mket_basis 7
mketp 6
mtrans 12

N

norm 9

O

op_trans 13

Q

qm_atrace 27
qm_mtrace 26
qm_variance 13

R

RX 14
RY 14
RZ 14

S

sigmax ... 10
sigmay ... 10
sigmaz ... 10
SM ... 13
spin_mbra ... 15
spin_mket ... 15
SP ... 13
stationary ... 25
Sx ... 11
SX ... 11
Sy ... 11
SY ... 12
Sz ... 11
SZ ... 12

T

tpbra ... 17
tpbraket ... 17
tpdagger ... 18
tpket ... 17

hbar ... 5

U

U ... 14
Uop ... 29

X

xm ... 9
xp ... 9

Y

ym ... 10
yp ... 10

Z

zm ... 9
zp ... 9